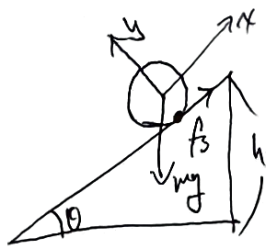


$$\boxed{2} \quad (1) \quad I_{\text{com}} + mr^2 = I_p$$

$$\begin{aligned} I_{\text{com}} &= I_p - mr^2 \\ &= \frac{3}{2}mr^2 - mr^2 \\ &= \frac{1}{2}mr^2 \end{aligned}$$

(2)



$$F_x: f_s - mg \sin \theta = m a_{\text{com}} \quad - (1)$$

$$F_y: -mg \cos \theta + N = 0 \quad - (2)$$

$$\tau_z: r \cdot f_s = I \alpha \quad - (3)$$

(3) or  $\alpha = -\frac{a_{\text{com}}}{r}$ ,  $\Rightarrow f_s = -\frac{I}{r^2} a_{\text{com}}$  or  $\text{neg.}$

$$-\frac{I}{r^2} a_{\text{com}} - mg \sin \theta = m a_{\text{com}}$$

$$a_{\text{com}} = \frac{-g \sin \theta}{1 + \frac{I}{mr^2}}$$

$$v = \int a dt = \frac{-g \sin \theta}{1 + \frac{I}{mr^2}} \cdot t \Rightarrow |v| = \frac{g \sin \theta}{1 + \frac{I}{mr^2}} t$$

$$s = \int |v| dt = \frac{1}{2} \cdot \frac{g \sin \theta}{1 + \frac{I}{mr^2}} t^2$$

$$s = \frac{h}{\sin \theta} \quad \text{or } (2/3)$$

$$\frac{1}{2} \cdot \frac{g \sin \theta}{1 + \frac{I}{mr^2}} t^2 = \frac{h}{\sin \theta}$$

$$t^2 = \frac{2h(1 + \frac{I}{mr^2})}{g \sin^2 \theta}$$

$$t = \frac{1}{\sin \theta} \cdot \sqrt{\frac{2h(1 + \frac{I}{mr^2})}{g}}$$

$$v_t = \frac{g \sin \theta}{(1 + \frac{I}{mr^2})} \cdot \frac{1}{\sin \theta} \sqrt{\frac{2h(1 + \frac{I}{mr^2})}{g}} = \sqrt{\frac{2gh}{1 + \frac{I}{mr^2}}} \quad - (4)$$

$$I = \frac{1}{2}mr^2 \quad \text{or } (2/3)$$

$$v_t = \sqrt{\frac{2gh}{1 + \frac{\frac{1}{2}mr^2}{mr^2}}} = \sqrt{\frac{4gh}{3}} \quad "$$